

Enzyme Action: Preparing the pH Investigation

I can explain enzyme specificity and denaturation, then identify the variables and evidence needed to investigate the effect of pH on amylase safely.

ACTIVE SITE

specific fit

CHANGE pH

one variable

TIME ENDPOINT

observable evidence

[Open the original interactive lesson](#)

What success looks like

- Use enzyme, substrate, active site and specificity accurately in an explanation.
- Explain how an unsuitable pH can change an active site's shape and reduce enzyme activity.
- Identify pH as the independent variable, time to starch disappearance as the dependent variable, and suitable control variables.

Retrieve and reconnect

- State what a biological catalyst does.
- Name the substrate broken down by amylase.
- Explain why temperature and solution volumes should stay constant in a pH investigation.
- Lead-in: predict what an extreme pH might do to an enzyme's active site.

Three amylase mixtures contain the same starch and are kept at the same temperature. Which first prediction is scientifically testable?

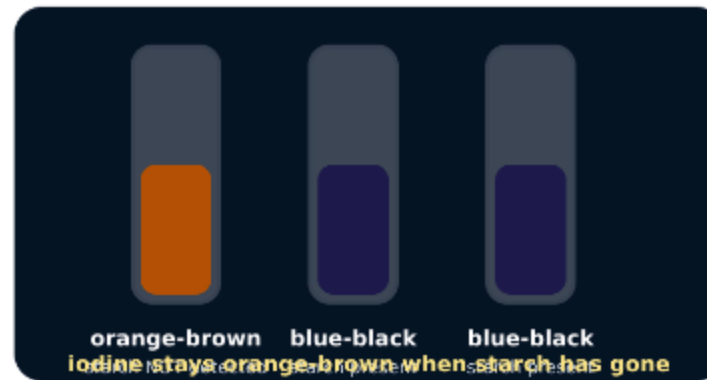
Think first. Point, say, sign or write your choice. Give one reason.

- A. Amylase will always work equally fast because enzymes never change.
- B. There may be an optimum pH where starch disappears in the shortest time.
- C. The buffer with the brightest label will make the reaction fastest.

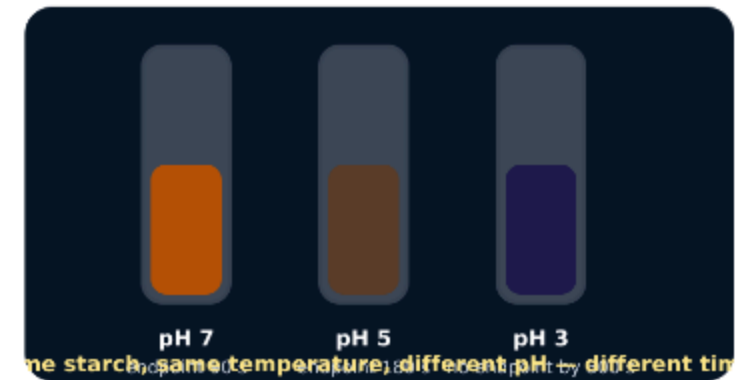
Look closely · what is the evidence?



Three tubes all blue-black with iodine at the start



One tube orange-brown, two still blue-black



Three tubes at different pH reaching the endpoint at different times

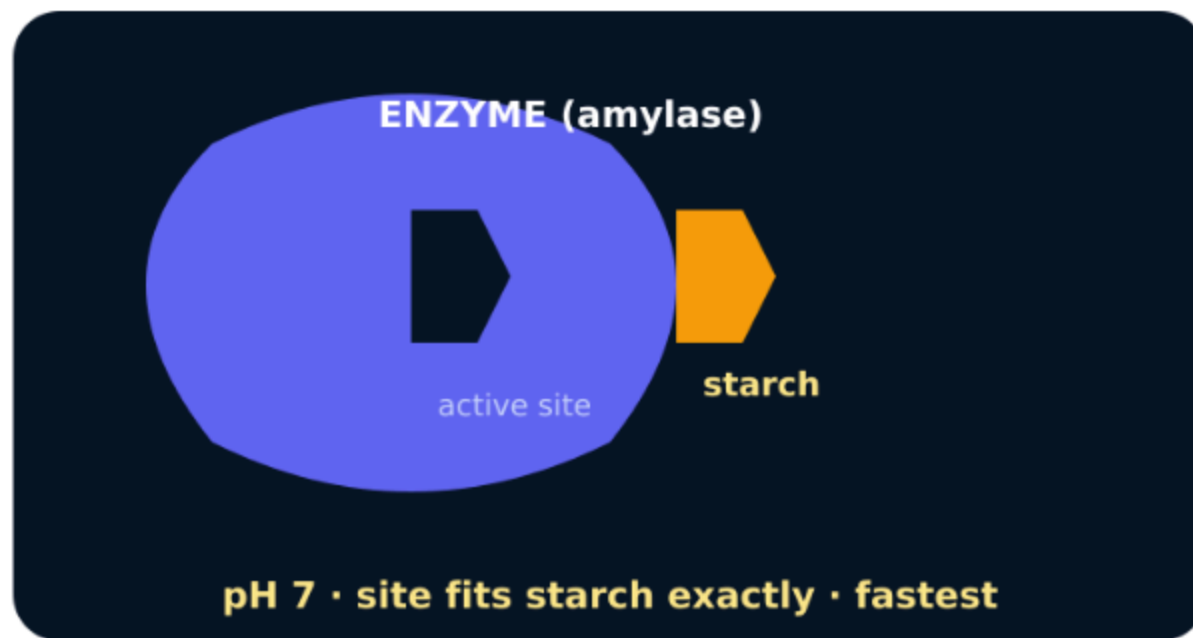
Words we will use

- specificity — one substrate fits
- denature — active site changes shape
- variable — what changes or is measured

Watch the model

- Represent amylase with one active-site shape and starch with a complementary substrate piece. Bind them briefly to show an enzyme-substrate complex.
- Separate the product pieces and return the enzyme model unchanged. The enzyme can catalyse another reaction and is not consumed.
- Alter the active-site model to represent denaturation. The substrate no longer fits well, so successful collisions and reaction rate fall.

Watch the model



Enzyme active site model at the chosen pH

Which statement best explains enzyme specificity?

- A. Every enzyme binds every substrate if enough time passes.
- B. An enzyme's active site has a complementary shape to its particular substrate.
- C. The substrate permanently changes the enzyme into a product.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Translate the model into a testable question: how does buffer pH affect the time amylase takes to digest starch?
- Set the variables: IV = buffer pH; DV = time until a sample no longer turns iodine blue-black; CVs include temperature, enzyme and starch concentrations and volumes, sampling interval and mixing method.
- Add a negative control sample with starch but no active amylase. It should continue to test positive for starch and helps show that starch loss depends on the enzyme.

Find and repair the misconception

Planning glitch: change pH, temperature and amylase volume together so the result is more dramatic.

A. Keep it: several changes make the evidence stronger.

B. Repair it: change pH only and control other rate factors so the comparison can test pH.

C. Repair it: remove the time measurement and judge by which tube looks best.

Variables and validity planning console

Route each planning card to independent variable, dependent evidence or control/validity before writing the method.

- Read one planning card.
- Commit it to IV, DV or control/validity.
- Repair any mismatch using the question being tested.
- Freeze the plan only when the cause, outcome and fair-comparison conditions are visible.

Variables and validity planning console

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- INDEPENDENT VARIABLE
- DEPENDENT EVIDENCE
- CONTROL / VALIDITY

State the predicted causal chain: changing pH may change active-site shape, which changes endpoint time and calculated rate.

Your independent mission

Produce a safe, ordered CP 1.10 planning record that another trained person could follow using the centre-approved Pearson method and local risk assessment.

Record: A pupil-owned prediction, IV/DV/CV table, ordered method outline, results-table headings and one justified safety/control decision.

Choose your support and challenge

- Supported: Complete a labelled IV/DV/CV table and order six picture-method cards; finish one because sentence about keeping temperature constant.
- Standard: Write an ordered method outline, identify variables and control sample, and design a table for pH, endpoint time and rate.
- Stretch: Justify the controls and sampling interval, predict the rate pattern, and identify one source of uncertainty before practical work begins.

Show what you know

Explain why temperature and volumes must be controlled when investigating pH and amylase.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.